

$$\begin{array}{l}
 \left[\begin{array}{ccc|ccc}
 2 & -6 & 6 & & & \\
 0 & -7 & 5 & & & \\
 0 & 8 & -4 & & & \\
 0 & 14 & 10 & & & \\
 0 & 21 & -15 & & &
 \end{array} \right] \xrightarrow{\substack{\text{Row3}=\text{Row3} \\ +8 \times \text{Row2}}} \left[\begin{array}{ccc|ccc}
 2 & -6 & 6 & & & \\
 0 & -7 & 5 & & & \\
 0 & 0 & 12/7 & & & \\
 0 & -14 & 10 & & & \\
 0 & 21 & -15 & & &
 \end{array} \right] \xrightarrow{\substack{\text{Row4}=\text{Row4} \\ -2 \times \text{Row2}}} \left[\begin{array}{ccc|ccc}
 2 & -6 & 6 & & & \\
 0 & -7 & 5 & & & \\
 0 & 0 & 12/7 & & & \\
 0 & 0 & 0 & & & \\
 0 & 0 & 0 & & &
 \end{array} \right] \\
 \xrightarrow{\substack{\text{Row5}=\text{Row5} \\ +3 \times \text{Row2}}} \left[\begin{array}{ccc|ccc}
 2 & -6 & 6 & & & \\
 0 & -7 & 5 & & & \\
 0 & 0 & 12/7 & & & \\
 0 & 0 & 0 & & & \\
 0 & 0 & 0 & & &
 \end{array} \right] \\
 L = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 & 0 \\ 3/2 & -8/7 & 1 & 0 & 0 \\ -3 & 2 & 0 & 1 & 0 \\ 4 & -3 & 0 & 0 & 1 \end{bmatrix} \\
 U = \begin{bmatrix} 2 & -6 & 6 & & & \\ 0 & -7 & 5 & & & \\ 0 & 0 & 12/7 & & & \\ 0 & 0 & 0 & & & \\ 0 & 0 & 0 & & & \end{bmatrix} \square
 \end{array}$$

17) When A is invertible, MATLAB finds A^{-1} by factoring $A=LU$ (where L may be permuted lower triangular), inverting L and U , and then computing $U^{-1}L^{-1}$. Use this method to compute the inverse of A in Exercise 2. (Apply the algorithm of Section 2.2 to L and to U .)

Solution:

$$A = \begin{bmatrix} 4 & 3 & -5 \\ -4 & -5 & 7 \\ 8 & 6 & -8 \end{bmatrix}$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}$$

$$U = \begin{bmatrix} 4 & 3 & -5 \\ 0 & -2 & 2 \\ 0 & 0 & 2 \end{bmatrix}$$

$$[U | I] \rightarrow [I | U^{-1}]$$

$$\left[\begin{array}{ccc|ccc}
 4 & 3 & -5 & 1 & 0 & 0 \\
 0 & -2 & 2 & 0 & 1 & 0 \\
 0 & 0 & 1 & 0 & 0 & 1/2
 \end{array} \right] \xrightarrow{\substack{\text{Row1}=\text{Row1} \\ +5 \times \text{Row3}}} \left[\begin{array}{ccc|ccc}
 4 & 3 & 0 & 1 & 0 & 5/2 \\
 0 & -2 & 2 & 0 & 1 & 0 \\
 0 & 0 & 1 & 0 & 0 & 1/2
 \end{array} \right]$$

$$\left[\begin{array}{ccc|ccc}
 4 & 3 & -5 & 1 & 0 & 0 \\
 0 & -2 & 2 & 0 & 1 & 0 \\
 0 & 0 & 2 & 0 & 0 & 1
 \end{array} \right] \xrightarrow{\substack{\text{Row3}=\text{Row3} \\ \times 1/2}} \left[\begin{array}{ccc|ccc}
 4 & 3 & -5 & 1 & 0 & 0 \\
 0 & -2 & 2 & 0 & 1 & 0 \\
 0 & 0 & 1 & 0 & 0 & 1/2
 \end{array} \right]$$

$$\xrightarrow{\times 1/2}$$

$$\xrightarrow{\substack{\text{Row2}=\text{Row2} \\ -2 \times \text{Row3}}} \left[\begin{array}{ccc|ccc}
 4 & 3 & 0 & 1 & 0 & 5/2 \\
 0 & -2 & 0 & 0 & 1 & -1/2 \\
 0 & 0 & 1 & 0 & 0 & 1/2
 \end{array} \right]$$

$$\xrightarrow{\substack{\text{Row2}=\text{Row2} \\ \times (-1/2)}} \left[\begin{array}{ccc|ccc}
 4 & 3 & 0 & 1 & 0 & 5/2 \\
 0 & 1 & 0 & 0 & -1/2 & 1/2 \\
 0 & 0 & 1 & 0 & 0 & 1/2
 \end{array} \right]$$

$$\left[\begin{array}{ccc|ccc}
 4 & 3 & 0 & 1 & 0 & 5/2 \\
 0 & -2 & 2 & 0 & 1 & 0 \\
 0 & 0 & 1 & 0 & 0 & 1/2
 \end{array} \right] \xrightarrow{\substack{\text{Row1}=\text{Row1} \\ -3 \times \text{Row2}}} \left[\begin{array}{ccc|ccc}
 4 & 0 & 0 & 1 & 3/2 & 1 \\
 0 & -2 & 2 & 0 & 1 & 0 \\
 0 & 0 & 1 & 0 & 0 & 1/2
 \end{array} \right]$$

$$\left[\begin{array}{ccc|ccc}
 4 & 3 & 0 & 1 & 0 & 5/2 \\
 0 & 1 & 0 & 0 & -1/2 & 1/2 \\
 0 & 0 & 1 & 0 & 0 & 1/2
 \end{array} \right] \xrightarrow{\substack{\text{Row1}=\text{Row1} \\ \times 1/4}} \left[\begin{array}{ccc|ccc}
 1 & 3/4 & 0 & 1/4 & 0 & 5/8 \\
 0 & 1 & 0 & 0 & -1/2 & 1/2 \\
 0 & 0 & 1 & 0 & 0 & 1/2
 \end{array} \right]$$

$$\xrightarrow{\substack{\text{Row1}=\text{Row1} \\ -3 \times \text{Row2}}} \left[\begin{array}{ccc|ccc}
 1 & 0 & 0 & 1/4 & 3/2 & 1/8 \\
 0 & 1 & 0 & 0 & -1/2 & 1/2 \\
 0 & 0 & 1 & 0 & 0 & 1/2
 \end{array} \right]$$

$$\left[\begin{array}{ccc|ccc}
 4 & 0 & 0 & 1 & 3/2 & 1 \\
 0 & -2 & 2 & 0 & 1 & 0 \\
 0 & 0 & 1 & 0 & 0 & 1/2
 \end{array} \right] \xrightarrow{\substack{\text{Row1}=\text{Row1} \\ \times 1/4}} \left[\begin{array}{ccc|ccc}
 1 & 0 & 0 & 1/4 & 3/2 & 1/8 \\
 0 & -2 & 2 & 0 & 1 & 0 \\
 0 & 0 & 1 & 0 & 0 & 1/2
 \end{array} \right]$$

$$\xrightarrow{\substack{\text{Row1}=\text{Row1} \\ \times 1/4}}$$